

CST444	SOFT COMPUTING	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PEC	2	1	0	3	2019

Preamble: This course enables the learners to understand the concepts of Soft Computing techniques and its applications. It covers Artificial Neural Networks, operations and models of fuzzy logic, genetic algorithms and multi objective optimization techniques. This course helps the students to develop algorithms and solutions for different real world applications.

Prerequisite: NIL.

Mapping of course outcomes with program outcomes

CO1	Describe soft computing techniques and the basic models of Artificial Neural Network (Cognitive Knowledge Level: Understand)
CO2	Solve practical problems using neural networks (Cognitive Knowledge Level: Apply)
CO3	Illustrate the operations, model and applications of fuzzy logic (Cognitive Knowledge Level: Apply)
CO4	Illustrate the concepts of Genetic Algorithm (Cognitive Knowledge Level: Apply)
CO5	Describe the concepts of multi-objective optimization models and the need for using hybrid soft computing approaches (Cognitive Knowledge Level: Understand)

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

Abstract POs defined by National Board of Accreditation			
PO#	Broad PO	PO#	Broad PO
PO1	Engineering Knowledge	PO7	Environment and Sustainability
PO2	Problem Analysis	PO8	Ethics
PO3	Design/Development of solutions	PO9	Individual and team work
PO4	Conduct investigations of complex problems	PO10	Communication
PO5	Modern tool usage	PO11	Project Management and Finance
PO6	The Engineer and Society	PO12	Life long learning

Assessment Pattern

Bloom's Category	Continuous Assessment Tests		End Semester Examination Marks (%)
	Test 1 (%)	Test 2 (%)	
Remember	30	30	30
Understand	30	30	30
Apply	40	40	40
Analyze			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	50	100	3

Continuous Internal Evaluation Pattern:

Attendance	10 marks
Continuous Assessment Tests(Average of Internal Tests 1 & 2)	25 marks
Continuous Assessment Assignment	15 marks

Internal Examination Pattern

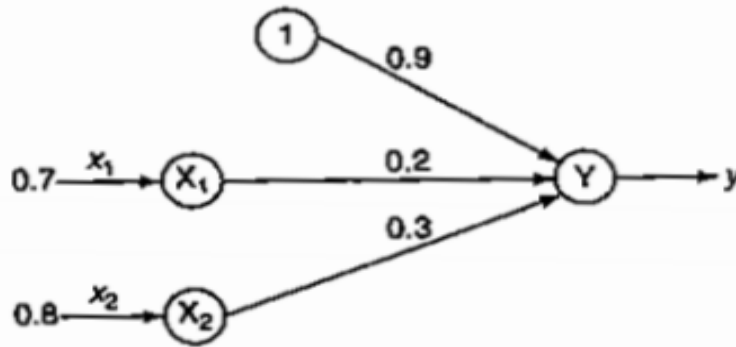
Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern:

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 full questions from each module of which student should answer any one. Each question can have maximum 2 sub-divisions and carries 14 marks.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Describe the necessity of Activation function? Examine the various aspects of sigmoidal activation function. List the drawbacks. Calculate the net output of the following neural network using the bipolar and binary sigmoidal activation function.



2. Explain the architecture of McCulloch-Pitts Neuron network model. Implement NAND(NOT-AND) gate function using M-P Neuron Model(with binary input).

Course Outcome 2(CO2):

1. Find the weights required to perform classification of patterns shown below using perceptron network. The patterns (1,1,-1) and (1, -1,-1) are belonging to the target class -1. The patterns (-1,1,1) and (-1,-1,1) are belonging to the target class +1. Assume suitable learning rate and initial weights.
2. Explain the architecture and training algorithm of Adaline network . Use Adaline network to train NOR logic function with bipolar inputs and targets. Perform 2 epochs of training.

Course Outcome 3(CO3):

1. There is an imprecise relationship between the ambient temperature for clay masonry bricks and their compressive strengths. Let X be a fuzzy set of fracture strengths and Y be a fuzzy set of temperatures with the following membership functions:

$$X = \left\{ \frac{1.0}{1500} + \frac{0.8}{2175} + \frac{0.6}{7000} + \frac{0.5}{12750} + \frac{0.3}{16500} + \frac{0.1}{20000} \right\}$$

$$Y = \left\{ \frac{0.2}{20} + \frac{0.4}{25} + \frac{0.5}{32} + \frac{1.0}{50} + \frac{0.6}{90} + \frac{0.3}{105} \right\}$$

- (a) Find the Cartesian Product of X and Y and represent it as relation R.
Suppose there is a second fuzzy set of masonry lengths given as

$$Z = \left\{ \frac{0.4}{1500} + \frac{0.5}{2175} + \frac{0.6}{7000} + \frac{0.8}{12750} + \frac{0.9}{16500} + \frac{1.0}{20000} \right\}$$

- (b) Find S=ZoR using max-min composition (c) Find T=ZoR using max-product composition

2. Given two universes $X=\{x_1, x_2, x_3, x_4, x_5\}$ and $Y=\{y_1, y_2, y_3, y_4, y_5\}$, the fuzzy sets A defined on X and fuzzy set B defined on Y are given below:

$$A = \left\{ \frac{0.4}{x_1} + \frac{0.7}{x_2} + \frac{1}{x_3} + \frac{0.8}{x_4} + \frac{0.6}{x_5} \right\} \quad B = \left\{ \frac{0.2}{y_1} + \frac{0.6}{y_2} + \frac{1}{y_3} + \frac{0.9}{y_4} + \frac{0.7}{y_5} \right\}$$

(i) Find the relation $R = A \times B$

Consider another fuzzy set C defined on the universe $V=\{v_1, v_2, v_3\}$, $C = \left\{ \frac{0.4}{v_1} + \frac{1}{v_2} + \frac{0.8}{v_3} \right\}$

(ii) Find $P = B \times C$. Using max-min composition, Find RoP .

Course Outcome 4(CO4):

1. Illustrate the various types of cross over with suitable examples.
2. Using Genetic algorithm with Roulette wheel selection method maximize the function $f(x)=x^2$ over $\{0, 1, 2, \dots, 31\}$ with initial x values of (13, 24, 8, 19). Show one crossover and mutation.

Course Outcome 5(CO5):

1. Explain strong dominance and weak pareto-optimality.
2. What are the different classifications of neuro-fuzzy hybrid systems?

Syllabus

Module – 1 (Introduction to Soft Computing & Artificial Neural Network)

Introduction to Soft Computing. Difference between Hard Computing & Soft Computing. Applications of Soft Computing. Artificial Neurons Vs Biological Neurons. Basic models of artificial neural networks – Connections, Learning, Activation Functions. McCulloch and Pitts Neuron. Hebb network.

Module – 2 (Supervised Learning Network)

Perceptron Networks– Learning rule, Training and testing algorithm. Adaptive Linear Neuron– Architecture, Training and testing algorithm. Back propagation Network – Architecture, Training and testing algorithm.

Module - 3 (Fuzzy Logic & Defuzzification)

Fuzzy sets – properties, operations on fuzzy set. Fuzzy membership functions, Methods of membership value assignments – intuition, inference, Rank Ordering. Fuzzy relations– operations on fuzzy relation. Fuzzy Propositions. Fuzzy implications. Defuzzification– Lamda cuts, Defuzzification methods.

Module - 4 (Fuzzy Inference System & Genetic Algorithm)

Fuzzy Inference Systems - Mamdani and Sugeno types. Fuzzy Logic Controller. Concepts of genetic algorithm. Operators in genetic algorithm - coding, selection, cross over, mutation. Stopping condition for genetic algorithm.

Module - 5 (Multi Objective Optimization & Hybrid Systems)

Multi objective optimization problem. Principles of Multi- objective optimization, Dominance and pareto-optimality. Optimality conditions. Neuro-fuzzy hybrid systems. Genetic – neuro hybrid systems.

Text Books

1. S.N.Sivanandam and S.N. Deepa, Principles of Soft Computing , 2nd Edition, John Wiley & Sons.
2. Kalyanmoy Deb, Multi-objective Optimization using Evolutionary Algorithms, 1st Edition, John Wiley & Sons.

Reference Books

1. Timothy J Ross, Fuzzy Logic with Engineering Applications, John Wiley & Sons, 2016.
2. T.S.Rajasekaran, G.A.Vijayalakshmi Pai “Neural Networks, Fuzzy Logic & Genetic Algorithms Synthesis and Applications”, Prentice-Hall India.
3. Simon Haykin, “Neural Networks- A Comprehensive Foundation”, 2/e, Pearson Education.
4. Zimmermann H. J, “Fuzzy Set Theory & Its Applications”, Allied Publishers Ltd.

Model Question Paper**QP CODE:****Reg No:** _____**Name:** _____**PAGES : 4****APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR****Course Code: CST 444****Course Name: Soft Computing****Max. Marks : 100****Duration: 3 Hours****PART A****Answer All Questions. Each Question Carries 3 Marks**

1. Explain the architecture of a simple Artificial Neural network? Compare it with a biological neuron.
2. A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. Predict the output?
3. Explain the Widrow-Hoff learning rule for supervised learning in neural networks with help of an example. Why is it sometimes called the LMS learning rule?
4. Implement one epoch of Adaline algorithm for AND logic function with binary inputs and bipolar outputs. Initial weights are $w_1=0.2$, $w_2=0.1$ and learning rate parameter $\eta=0.2$.
5. Consider two fuzzy sets $A = \left\{ \frac{0.2}{0} + \frac{0.3}{1} + \frac{1}{2} + \frac{0.1}{3} + \frac{0.5}{4} \right\}$ $B = \left\{ \frac{0.1}{0} + \frac{0.25}{1} + \frac{0.9}{2} + \frac{0.7}{3} + \frac{0.3}{4} \right\}$ Find the following: (a) Algebraic sum (b) Algebraic product (c) Bounded sum.
6. Using your own intuition and definition of universe of discourse, plot membership

functions for liquid level (Empty, very less, less, full, very full) in a tank.

7. Explain Stochastic Universal Sampling with an example.
8. Explain any two mutation methods.
9. Differentiate between linear and nonlinear Multi Objective Optimization Problem.
10. What are the characteristics of neuro fuzzy hybrid systems?

(10x3=30)

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Implement XOR function using M-P Neuron Model (with binary input). Why M-P neuron is widely used in processing binary data? (8)
 - (b) Using Hebb Network calculate the weight required to perform the following classification of given input pattern. (6)
- L $\square$ belongs to the members of the class(+) $\square$ target value +1
 U $\square$ does not belongs to members of class(.) $\square$ target value -1

+	.	.
+	.	.
+	+	+

L

+	.	+
+	.	+
+	+	+

U

OR

12. (a) Compare the three learning approaches in Artificial Neural Network. How is the critic information used in learning process. (8)
- (b) Define Hebb Law. Design a Hebb Network to implement logical AND function. Use bipolar input and targets. (7)

13. (a) Discuss the training algorithm and explain the weight updates in back propagation networks. (10)

(b) Implement one epoch of Perceptron training algorithm for OR logic function with binary input and bipolar output. (4)

OR

14. (a) Explain how synaptic weights are adapted iteration by iteration using error correction rule in Perceptron convergence algorithm for an OR gate with bipolar inputs and outputs. Initial weights are all zero and learning rate parameter $\eta=0.1$. (10)

(b) Explain Perceptron convergence theorem and discuss Perceptron algorithm based on XOR logic function. (4)

15. (a) Three fuzzy sets are defined as follows: (10)

$$A = \left\{ \frac{0.1}{30} + \frac{0.2}{60} + \frac{0.3}{90} + \frac{0.4}{120} \right\}, B = \left\{ \frac{1}{1} + \frac{0.2}{2} + \frac{0.5}{3} + \frac{0.7}{4} + \frac{0.3}{5} + \frac{0}{6} \right\},$$

$$C = \left\{ \frac{0.33}{100} + \frac{0.65}{200} + \frac{0.92}{300} + \frac{0.21}{400} \right\}$$

Find: (i) $R = A \times B$ (ii) $S = B \times C$ (iii) $T = RoS$, using Max-Min composition
(iv) $T = RoS$, using Max-Product composition.

(b) For the fuzzy sets given $A = \left\{ \frac{0.5}{x_1} + \frac{0.2}{x_2} + \frac{0.9}{x_3} \right\}$ and $B = \left\{ \frac{1}{y_1} + \frac{0.5}{y_2} + \frac{1}{y_3} \right\}$. Find relation R by performing Cartesian product over the given fuzzy sets. (4)

OR

16. (a) Using inference approach, find the membership values for each of the triangular shapes (I, R, IR, T) for a triangle with angles 120° , 50° , 10° . (8)

(b) Using Zadeh's notation, determine the λ - cut sets for the given fuzzy sets: (6)

$$S_1 = \left\{ \frac{0}{0} + \frac{0.5}{20} + \frac{0.65}{40} + \frac{0.85}{60} + \frac{1.0}{80} + \frac{1.0}{100} \right\}$$

$$S_2 = \left\{ \frac{0}{0} + \frac{0.45}{20} + \frac{0.6}{40} + \frac{0.8}{60} + \frac{0.95}{80} + \frac{1.0}{100} \right\}$$

Express the following for $\lambda = 0.5$: a) $S_1 \cup S_2$ b) S_2' c) $S_1 \hat{\cap} S_2$

17. (a) Differentiate between value encoding and permutation encoding. (8)

(b) Explain the stopping conditions for genetic algorithm. (6)

OR

18. (a) Apply Mamdani fuzzy model to design a controller to determine the wash time of a domestic washing machine. Assume input is dirt and grease of the cloth. Use three descriptors for input variable and five descriptors for output variables. Derive the set of rules for controller action and defuzzification. Design should be supported by figure wherever possible. (10)

(b) Explain Single-Point Crossover and Two-Point Crossover with example. (4)

19. (a) Explain convex and non convex MOOP? How to find a non dominated set. (10)

(b) What are the properties of dominance relation? (4)

OR

20. (a) Explain Genetic Neuro-Hybrid System with block diagram. Also write the advantages of Genetic- Neuro Hybrid systems. (8)

(b) Discuss the classification of Neuro-Fuzzy Hybrid System. (6)

Teaching Plan

No	Contents	No. of Lecture Hours (35 hrs)
Module-1 (Introduction to Soft Computing & Artificial Neural Network) (6 hours)		
1.1	Introduction to Soft Computing	1 hour
1.2	Difference between Hard Computing & Soft Computing & Applications of Soft Computing	1 hour
1.3	Artificial Neurons Vs Biological Neurons, Basic models of artificial neural networks	1 hour
1.4	Activation Functions	1 hour
1.5	McCulloch and Pitts Neuron	1 hour
1.6	Hebb network	1 hour
Module-2 (Supervised Learning Network) (7 hours)		
2.1	Perceptron networks – Learning rule, Training and testing algorithm	1 hour
2.2	Perceptron networks – Problems	1 hour
2.3	Adaptive Linear Neuron (Lecture I)	1 hour
2.4	Adaptive Linear Neuron (Lecture II)	1 hour
2.5	Adaptive Linear Neuron-Problems (Lecture III)	1 hour
2.6	Back propagation Network (Lecture I)	1 hour
2.7	Back propagation Network (Lecture II)	1 hour
Module-3 (Fuzzy Logic & Defuzzification) (8 hours)		
3.1	Introduction to Fuzzy Set, Properties & operations on fuzzy sets	1 hour
3.2	Fuzzy membership functions, Fuzzification	1 hour
3.3	Methods of membership value assignments	1 hour
3.4	Fuzzy relations, Operations on Fuzzy Relation	1 hour

3.5	Fuzzy Propositions & Fuzzy Implications	1 hour
3.6	Lamda cuts for fuzzy sets	1 hour
3.7	Defuzzification methods(Lecture I)	1 hour
3.8	Defuzzification methods(Lecture II)	1 hour
Module-4 (Fuzzy Inference System & Genetic Algorithm) (6 hours)		
4.1	Fuzzy Inference Systems - Mamdani type	1 hour
4.2	Fuzzy Inference Systems - Sugeno type	1 hour
4.3	Fuzzy Logic Controller	1 hour
4.4	Introduction to genetic algorithm, operators in genetic algorithm - coding	1 hour
4.5	Selection, Cross over	1 hour
4.6	Mutation, stopping condition for genetic algorithm	1 hour
Module-5 (Multi-Objective Optimization & Hybrid System) (8 hours)		
5.1	MOOP-Linear & Non linear, Convex & Non Convex	1 hour
5.2	Principles of MOO-Illustrating Pareto Optimal Solutions, Objectives in MOO	1 hour
5.3	Dominance & Pareto-Optimality-Concept of Domination	1 hour
5.4	Properties of Dominance Relation, Pareto Optimality	1 hour
5.5	Procedure for finding a non dominated set	1 hour
5.6	Optimality Conditions	1 hour
5.7	Neuro Fuzzy hybrid system-Classification& characteristics	1 hour
5.8	Genetic –neuro hybrid systems	1 hour